

Campus Productivity Tool: Student Portal Management System

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Abstract—The Student Portal Management System is a role-based web application developed to manage and integrate academic, student, teacher, examination, and placement activities through a centralized platform. The system provides separate functionalities for Admin, Teacher, and Student users. Admins can manage student and teacher details, campus interview schedules, and company profiles with eligibility criteria based on 10th and 12th-grade percentages. Teachers can manage student timetables and attendance records, question papers and answers. Students can view their academic information, upload documents for RAG-based text summarization and MCQ generation, and upload resumes for company eligibility analysis. The system also provides email alerts for upcoming campus interviews and supports automated answer-sheet evaluation by comparing student answers with stored answers using similarity-based analysis. The answer analysis is expected to achieve approximately 80–90% effectiveness, depending on document quality and answer similarity. Overall, the system aims to simplify academic management, support AI-assisted learning, automate basic assessment processes, and improve campus placement coordination through a single intelligent web-based platform.

Keywords—Student Portal Management, Role-Based Access Control, Retrieval-Augmented Generation, Large Language Models, Text Summarization, MCQ Generation, Automated Answer Evaluation, Similarity Analysis, Campus Placement Management, Email Notification.

I. INTRODUCTION

A. General

The Student Portal Management System is a centralized role-based web application designed to integrate and simplify various academic and placement-related activities within an educational institution. The system provides separate functionalities for Admin, Teacher, and Student users, allowing each role to perform their respective activities through a unified platform. Admins can manage student and teacher information, company details, eligibility criteria, and campus interview

schedules, while teachers can maintain student timetables, attendance records, question papers, and answers. Students can access their academic information, upload study documents for RAG-based text summarization and MCQ generation, and upload resumes for automated company eligibility analysis. The system also supports email notifications for upcoming campus interviews and similarity-based automated answer-sheet evaluation. By combining conventional academic management features with AI-assisted functionalities, the proposed system provides an efficient and intelligent platform for managing student activities, supporting personalized learning, and improving placement coordination.

B. Scope of the Project

The scope of the Student Portal Management System covers the centralized management of academic, examination, learning, and placement activities in an educational institution. The system enables Admins to manage student and teacher profiles, company information, interview schedules, and eligibility requirements based on academic performance. Teachers can maintain student timetables and attendance, prepare question papers, and store correct answers for assessment purposes. Students can view their academic details and use AI-assisted learning features by uploading study materials for RAG-based text summarization and MCQ generation. The system also provides resume-based company eligibility analysis to help students identify suitable placement opportunities. Automated email alerts can notify students about upcoming campus interviews, while answer-sheet evaluation can compare student responses with stored answers using similarity-based analysis.

C. Objective

The main objective of the Student Portal Management System is to develop an integrated web-based platform

that simplifies academic management and enhances student learning and placement activities. The system aims to provide role-based access for Admin, Teacher, and Student users while allowing efficient management of academic and institutional information. Another objective is to support students with AI-based document summarization and MCQ generation using RAG techniques, enabling them to learn from their own study materials. The system also aims to automate basic examination activities through similarity-based answer-sheet evaluation and assist students in identifying companies for which they satisfy the required eligibility criteria. Additionally, the system provides timely email alerts for campus interviews, thereby reducing manual communication and improving placement coordination.

D. Motivation of the Project

The motivation for developing the Student Portal Management System comes from the need to reduce the dependency on multiple disconnected processes for managing student academic, examination, learning, and placement activities. In many educational environments, student information, attendance, timetables, examination materials, placement details, and interview notifications may be maintained through separate systems or manual procedures, making information management time-consuming and less efficient. Students may also face difficulties in understanding large study materials and identifying important topics for examination preparation. To address these challenges, the proposed system incorporates RAG-based text summarization and MCQ generation, allowing students to obtain useful learning content directly from their uploaded documents.

E. Problem Statement

Educational institutions often manage student academic information, attendance, timetables, examinations, learning materials, and placement activities through separate processes, which can result in increased manual effort, information duplication, and difficulty in accessing required information. Teachers may need to maintain student records and evaluate answer sheets manually, while students may have limited support for quickly understanding lengthy study materials and preparing practice questions from their own documents. Therefore, there is a need for a centralized Student Portal Management System that integrates academic and placement management with AI-assisted learning and automation features.

II. LITERATURE SURVEY

A. Exploring the Landscape of Automatic Text Summarization: A Comprehensive Survey

Bilal Khan, Zohaib Ali Shah, Muhammad Usman, Inayat Khan, and Badam Niazi (2023) present a comprehensive survey of Automatic Text Summarization (ATS), covering extractive, abstractive, and hybrid approaches. The survey examines challenges, classifications, datasets, and evaluation measures relevant to ATS research, providing a foundation for the RAG-based summarization approach used in this project.

B. Two-Stage Text Summary Model

Yang Chen and Roben A. Juanatas (2024) propose a two-stage text summarization model combining an abstraction stage with dilated and gated convolution and a generation stage using a replication mechanism. Evaluated on the CNewSum dataset, the model achieved improved ROUGE scores over benchmark models, demonstrating the value of structured multi-stage summarization pipelines.

C. MCQGen: A Large Language Model-Driven MCQ Generator for Personalized Learning

Ching Nam Hang, Chee Wei Tan, and Pei-Duo Yu (2024) introduce MCQGen, a generative AI framework that integrates an LLM with retrieval-augmented generation and prompt engineering techniques such as chain-of-thought and self-refine prompting to automatically generate contextually relevant multiple-choice questions, directly motivating the MCQ generation module of this project.

D. Assessing the Impact of Chatbot-Human Personality Congruence on User Behavior

Mohammad Amin Kuhail et al. (2024) study how personality congruence between chatbots and users affects trust and engagement in a chatbot-based advising system, offering insight relevant to designing user-facing AI features within an academic portal.

E. Data Augmentation and Preparation for PerInfEx: A Persian Chatbot with Information Extraction

Pegah Safari and Mehrnoush Shamsfard (2024) describe data preparation and augmentation techniques for a Persian information-extraction chatbot, evaluating NLU performance with intent-accuracy of 88.64%, slot-F1 of 83.68%, and exact-match of 78.22%, underscoring the importance of quality training data for NLP-driven system components.

F. Conclusion

The reviewed literature establishes that transformer-based summarization, retrieval-augmented generation, and LLM-driven question generation are mature and effective techniques for AI-assisted learning applications. This motivates their integration into the Student Portal Management System's document-processing module, alongside the platform's academic, examination, and placement management functions.

III. SYSTEM DESIGN

A. Existing System

The existing system for managing student-related activities is generally based on separate manual or basic digital processes for academic, examination, attendance, and placement activities. Student and teacher details may be maintained independently, while attendance, timetables, question papers, and answer sheets are often handled manually or through different applications. Placement information such as company profiles, eligibility criteria, and interview schedules may also be managed separately, requiring students and staff to track information manually. In such systems, there is limited integration of AI-based learning features, document

summarization, automatic MCQ generation, resume-based eligibility analysis, and automated answer evaluation.

1) Disadvantages

- Academic, examination, attendance, and placement activities are often managed through separate systems or manual processes.
- Maintaining student records, attendance, timetables, and examination details manually is time-consuming and may lead to errors.
- Students have limited support for quickly summarizing study materials and generating practice questions.
- Placement eligibility checking and campus interview notifications may require manual tracking, which can cause delays or missed opportunities.

B. Proposed System

The proposed Student Portal Management System is a role-based web application designed to manage academic, examination, and placement activities through a centralized platform. The system provides separate access for Admin, Teacher, and Student users. The Admin can manage student and teacher details, store corresponding answers, update campus interview schedules, and maintain company profiles with eligibility criteria based on 10th and 12th-grade percentages. Teachers can create and manage student timetables and update attendance records, and prepare question papers through the system. Students can view their timetable and attendance details and access various AI-based learning and placement services.

The proposed system integrates Retrieval-Augmented Generation (RAG) for document-based processing. Students can upload documents and generate meaningful text summaries and multiple-choice questions based on the uploaded content. The system also supports resume-based eligibility analysis by comparing students' academic percentages with company eligibility requirements stored by the Admin. Email notifications are provided to students before scheduled campus interview dates. In addition, students can upload test answer-sheet documents, which

are processed and compared with the correct answers stored in the database using similarity-based analysis. The system identifies correctly and incorrectly answered questions with an expected effectiveness of approximately 80–90%, depending on document quality and answer similarity.

1) Advantages

- Provides a centralized platform for managing student, teacher, examination, attendance, and placement activities, reducing manual work.
- Supports AI-based document processing to generate summaries and MCQs, helping students with personalized learning.
- Automates placement activities through resume-based eligibility checking and email notifications for campus interviews.
- Enables automated answer-sheet evaluation using similarity-based analysis, reducing manual correction time and improving evaluation efficiency.

C. System Requirements

The hardware and software requirements of the proposed system are listed in Table.1.

Component	Specification
Hardware Requirements	
Processor	Intel i3 / i5 / i7
RAM	8 GB
Hard Disk	500 GB
Software Requirements	
Operating System	Windows 8 / 10
Scripting Language	Python (3.10.8)
Development Tool	IDLE (Python)

Table.1.Hardware and Software Requirements

D. Architecture Diagram

The overall architecture of the Student Portal Management System is illustrated in Fig.1.

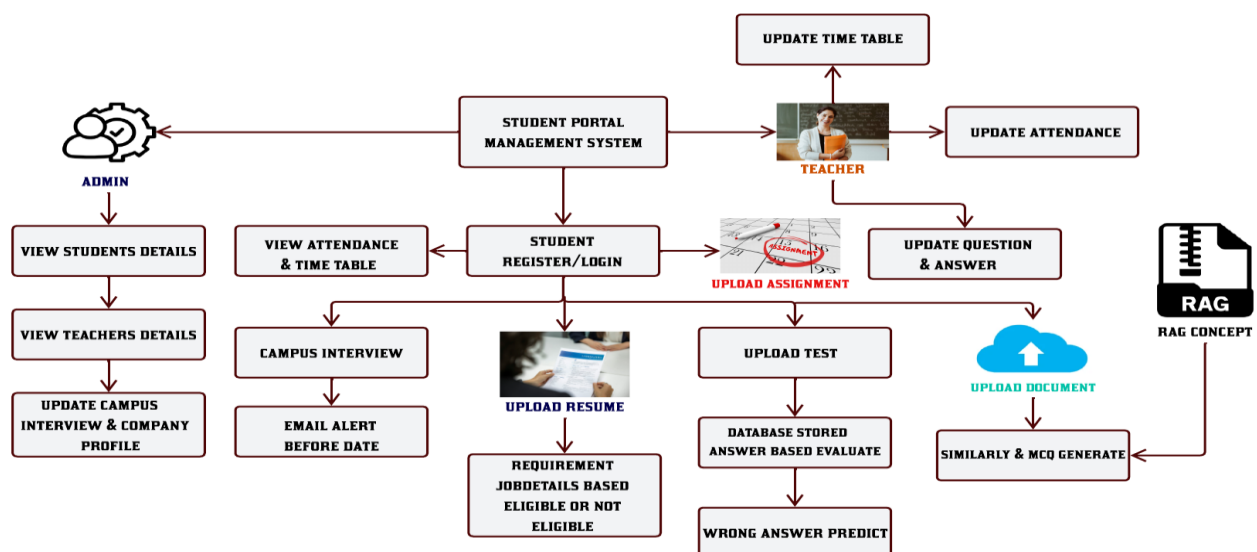


Fig.1. Architecture Diagram for the Student Portal Management System

IV. PROJECT DESCRIPTION

A. Modules

1) *Admin Module*: The Admin Module manages the overall system and maintains student and teacher details. Admin can manage campus interview schedules and company profiles. Company eligibility criteria based on 10th and 12th-grade percentages can be added and updated. The module also stores correct answers for examination and answer evaluation.

2) *Teacher Module*: The Teacher Module allows teachers to manage academic-related activities. Teachers can create and update student timetables and maintain attendance records. They can prepare question papers and store the corresponding answers in the system.

3) *Student Module*: The Student Module provides students with access to their academic and placement-related information. Students can view their timetable and attendance details. They can upload study documents for AI-based processing and generate summaries and MCQs. Students can also upload resumes for company eligibility analysis.

4) *AI-Based Document Processing Module*: The AI-Based Document Processing Module processes documents uploaded by students using RAG-based techniques. The system extracts meaningful information from the uploaded content and generates concise text summaries. It can also generate multiple-choice questions based on the document content.

5) *Examination & Answer Evaluation Module*: The Examination and Answer Evaluation Module manages question papers, correct answers, and student answer sheets. Students can upload their completed answer-sheet documents through the system. The uploaded answers are processed and compared with the correct answers stored in the database using similarity-based analysis. The expected effectiveness is approximately 80–90%, depending on document quality and answer similarity.

6) *Campus Placement Management Module*: The Campus Placement Management Module helps manage company profiles, eligibility criteria, and campus interview schedules. Students' academic percentages can be compared with the eligibility requirements of available companies. Email alerts are provided to students before scheduled campus interviews.

B. UML Diagrams

The Use Case Diagram, Activity Diagram, Data Flow Diagram, and Overall System Diagram of the proposed system are presented in Fig.2, Fig.3, Fig.4, and Fig.5 respectively.

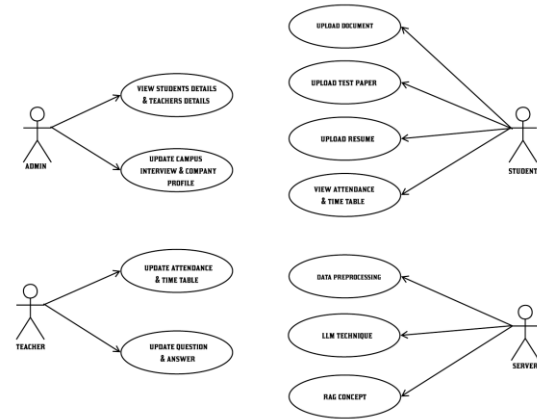


Fig.2. Use Case Diagram

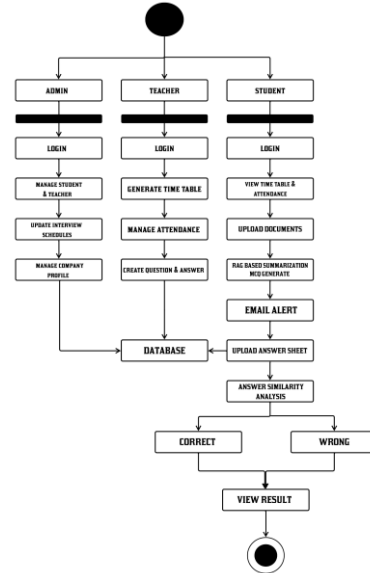


Fig.3. Activity Diagram

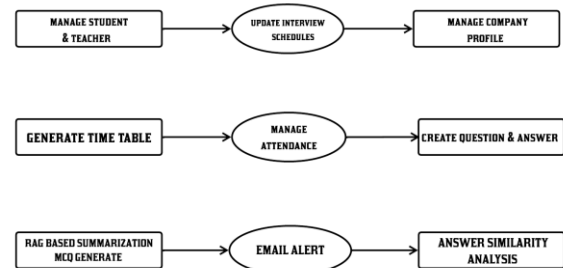


Fig.4. Data Flow Diagram

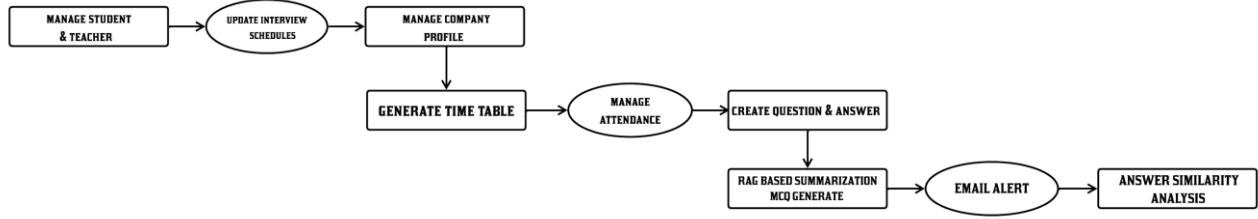


Fig.5. Overall System Diagram

C. Key Techniques Used

1) *Large Language Models (LLM)*: Large Language Models are deep neural network based AI systems that learn patterns, grammar, and context from text to answer questions, generate content, and support conversational interaction. The system draws on LLMs such as GPT and LLaMA-family models to power its document summarization and MCQ generation features.

2) *Retrieval-Augmented Generation (RAG)*: Retrieval-Augmented Generation combines information retrieval with an LLM to produce more accurate, context-relevant output. Uploaded documents are chunked, embedded, and stored for similarity-based retrieval; the most relevant chunks are then supplied as context to the LLM to generate summaries and MCQs grounded in the student's own material.

V. CONCLUSION

The proposed Student Portal Management System provides an integrated, role-based platform that combines academic administration, examination management, AI-assisted learning, and campus placement coordination. By incorporating RAG-based document summarization and MCQ generation, similarity-based automated answer evaluation, and resume-based eligibility analysis, the system reduces manual workload for administrators and teachers while giving students direct access to intelligent learning and placement support within a single web application.

VI. FUTURE WORK

Future enhancements can include more advanced NLP models to provide more accurate document summarization, MCQ generation, and answer evaluation. A chatbot can be added to help students with academic and placement-related queries. The placement module can be extended with job recommendation features based on students' skills, academic performance, and resumes. Mobile application support can also be introduced to provide students and teachers with easy access to the system from anywhere. In addition, advanced analytics and dashboards can be implemented to track student performance, attendance, examination results, and placement progress.

VII. REFERENCES

[1] T. Nguyen, Q. Pham, L. Chen, and R. Zhang, "A review on personalized learning and intelligent

tutoring systems using deep learning and graph-based approaches," *IEEE Access*, 2023.

- [2] A. A. Syed, F. L. Gaol, and T. Matsuo, "A survey of the state-of-the-art models in neural abstractive text summarization," *IEEE Access*, vol. 9, pp. 13248–13265, 2021.
- [3] A. Elsaid, A. Mohammed, L. F. Ibrahim, and M. M. Sakre, "A comprehensive review of Arabic text summarization," *IEEE Access*, vol. 10, pp. 38012–38030, 2022.
- [4] S. Tanwar, Q. Bhatia, P. Patel, A. Kumari, P. K. Singh, and W.-C. Hong, "Machine learning adoption in blockchain-based smart applications: The challenges, and a way forward," *IEEE Access*, vol. 8, pp. 474–488, 2020.
- [5] B. Khan, R. Naseem, F. Muhammad, G. Abbas, and S. Kim, "An empirical evaluation of machine learning techniques for chronic kidney disease prophecy," *IEEE Access*, vol. 8, pp. 55012–55022, 2020.
- [6] C. W. Tan, L. Ling, P.-D. Yu, C. N. Hang, and M. F. Wong, "Mathematics gamification in mobile app software for personalized learning at scale," *Proc. IEEE Integr. STEM Educ. Conf. (ISEC)*, Aug. 2020, pp. 1–5.
- [7] D. R. Ch and S. K. Saha, "Automatic multiple choice question generation from text: A survey," *IEEE Trans. Learn. Technol.*, vol. 13, no. 1, pp. 14–25, Jan. 2020.
- [8] Y. Chen, W. Huang, and K.-S. Lee, "Modeling knowledge acquisition with graph neural networks in educational systems," *IEEE Trans. Learn. Technol.*, 2024.
- [9] F. K. Sufi and I. Khalil, "Automated disaster monitoring from social media posts using AI-based location intelligence and sentiment analysis," *IEEE Trans. Computat. Social Syst.*, 2022.
- [10] R. Pramana, Debora, J. J. Subroto, A. A. S. Gunawan, and Anderies, "Systematic literature review of stemming and lemmatization performance for sentence similarity," *Proc. IEEE 7th Int. Conf. Inf. Technol. Digit. Appl. (ICITDA)*, Nov. 2022, pp. 1–6.